

Multi-Sensor Fusion Pipeline

Road Quality & Litter Monitoring

RT-DETR + IMU + GPS — Late Fusion Approach

Transfer Learning on Unannotated Real-World Data

→ Problem Statement

Roads and urban environments degrade quietly. Our sensor data is **real but unannotated**.

⊗ Our Approach

Transfer learning from public datasets → fine-tune → deploy late fusion on real streams.

Pipeline Stages:

1. Data sourcing & transfer learning
2. Vision model — RT-DETR
3. IMU anomaly detection
4. Late sensor fusion
5. GPS geo-tagging
 - ! Why it failed & next steps

The Problem

Two separate monitoring challenges:

≡ **Road Defects:** Cracks, Potholes

∅ **Litter:** Plastic, Bottles, Waste

Both detected from a moving vehicle using cameras.

! Key Constraint

Our **real-world dataset is unannotated** — we have images from the road but no ground-truth labels.

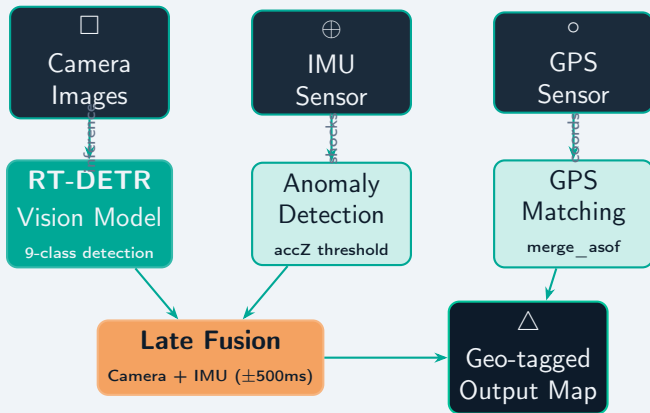
⦿ Solution

Use **transfer learning** from two public annotated datasets to build a model that generalises to our data.

Transfer Learning Strategy



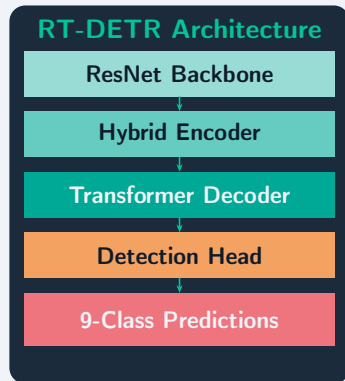
Full Pipeline Architecture



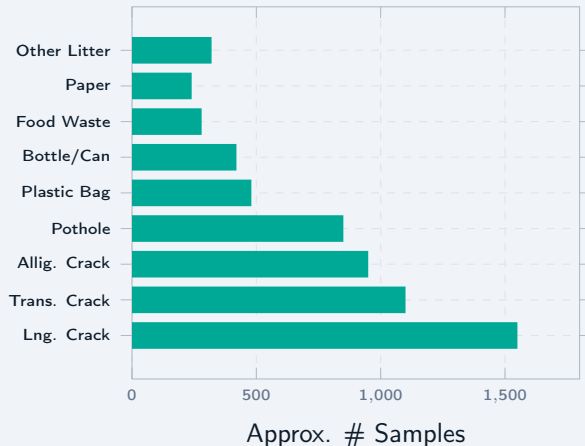
Vision Model — RT-DETR

Why RT-DETR?

- **Transformer-based** detector — no anchor boxes needed
- **Real-Time** capable — efficient hybrid encoder
- **Pre-trained weights** available (rtdetr-1.pt) for transfer
- Better at **multi-scale** detection than classic YOLO on small objects



Dataset: 9 Detection Classes



Road Damage (RDD2022):

- D00 Longitudinal Crack
- D10 Transverse Crack
- D20 Alligator Crack
- D40 Pothole

Litter (TACO):

- Plastic Bag
- Bottle / Can
- Food Waste
- Paper
- Other Litter

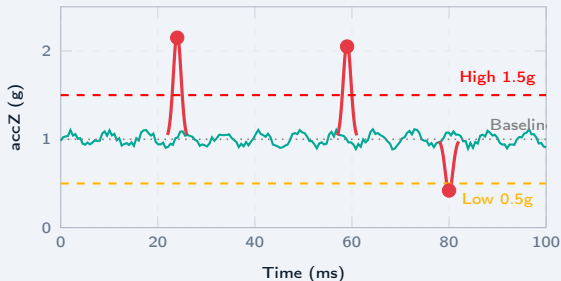
Note

Class imbalance: road damage $\gg$ litter. A key challenge for training stability.

IMU Anomaly Detection

Detecting shocks from accelerometer:

The vertical axis (accZ) at rest $\approx 1.0g$. A pothole causes a sharp deviation.



Detection Rule

$$\text{shock} = \begin{cases} \text{jolt_up} & \text{if } \text{accZ} > 1.5 \\ \text{drop_down} & \text{if } \text{accZ} < 0.5 \end{cases}$$

IMU stream columns:

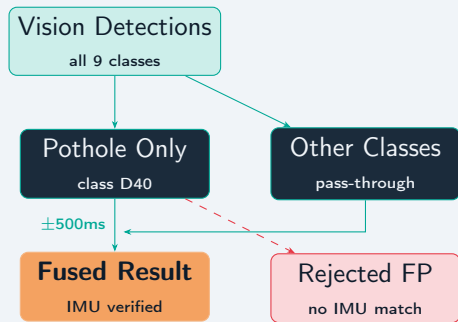
```
unixtime_ms, accX, accY,  
accZ, angX, angY, angZ,  
data_valid
```

✓ Only `data_valid == 1`

Late Fusion Strategy

Why Late Fusion?

Each sensor is processed *independently* first, then combined at decision level. No need to retrain on synchronized streams.



merge_asof Fusion

Logic: Only keep a pothole detection if there is an IMU shock within $\pm 500\text{ms}$.

$$|\Delta t| = |t_{\text{cam}} - t_{\text{imu}}| \leq 500 \text{ ms}$$

Intuition:

- Camera alone $\Rightarrow$ many false positives (shadows, markings)
- IMU alone $\Rightarrow$ no spatial / class context

GPS Geo-Tagging & Output Map

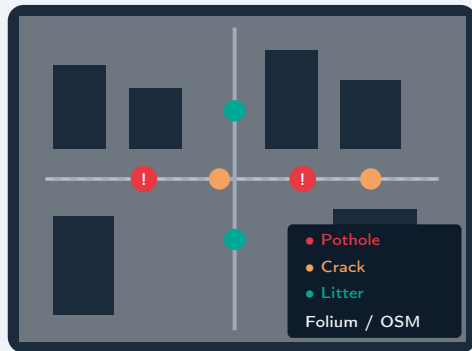
Frame timestamps are matched to GPS coordinates using `pandas.merge_asof` with a 1-second tolerance.

GPS columns:

`unixtime_ms`, `spd`, `lat`, `lon`

Output: Interactive Folium map with colour-coded markers:

Class	Color	Icon
Pothole	• Red	!
Crack types	• Orange	road
Litter	• Blue/Purple	trash



Why It Didn't Work — Domain Mismatch

Core Issue: Two Completely Different Visual Domains

RDD2022 images:

Overhead road shots, Japan/India
Clear lighting, dry roads
Dense crack annotations

! vs. TACO images:

Flickr / street photography
Variable angles, lighting, context
Litter in complex scenes

× What Happened

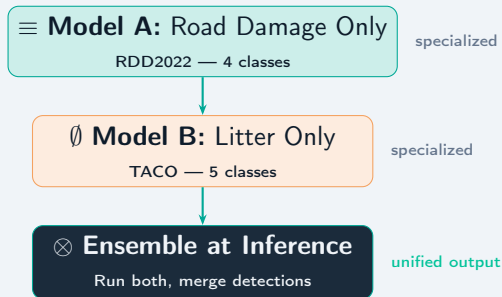
Training on the **merged** dataset caused the model to confuse features across domains. Crack textures interfered with litter classification and vice versa.

⦿ Root Cause

- Different scale of objects
- Different image statistics (mean, std)
- Class imbalance: 4 crack types vs 5 litter types — very skewed

Proposed Solution & Next Steps

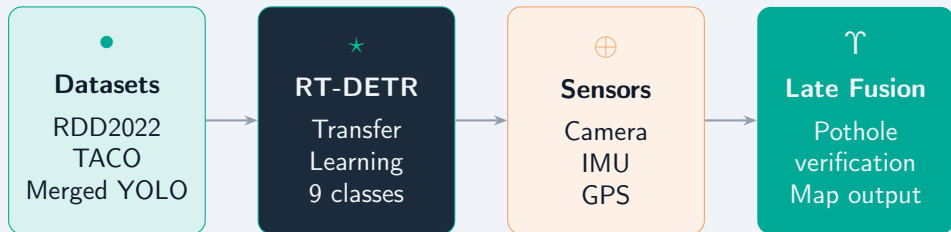
Separate → Train → Reassemble



Concrete next steps:

1. Train **Model A** on RDD2022 only (road damage, 4 classes)
2. Train **Model B** on TACO only (litter, 5 classes)
3. At inference: run both models, combine bounding boxes
4. Apply NMS across merged boxes
5. Feed combined output into existing late fusion pipeline

Summary



✓ What We Built

A full end-to-end pipeline: data ingestion, transfer learning, multi-sensor fusion, and interactive map output.

→ What Comes Next

Separate the two detection tasks into dedicated specialized models, then ensemble at inference time.

Thank You

Road Quality & Litter Monitoring

Next step: Separate models → Ensemble → Better precision